DECODELABS · INTERNSHIP SUBMISSION

Iris Flower Classification Using AI

An end-to-end supervised learning pipeline built as an AI Engineer project.

K-Nearest Neighbors · StandardScaler · Python

Verified test accuracy: 93.33%



PROJECT 02

Data Classification

2026 · Portfolio presentation

Problem & Objective

THE QUESTION

Can a machine learn to identify an Iris species from flower measurements?

Input: 4 numerical measurements
Output: 1 of 3 species classes

PROJECT OBJECTIVE

- Load and understand a labeled dataset
- Split data into training and testing sets
- Apply a simple supervised algorithm
- Validate the model with multiple metrics

Outcome: a reproducible, reviewable ML pipeline.

Dataset: Iris Benchmark

150
SAMPLES

4
FEATURES

3
CLASSES

0
MISSING VALUES

FEATURES USED

Sepal length	cm
Sepal width	cm
Petal length	cm
Petal width	cm

CLASS BALANCE



Balanced classes support fairer evaluation.

From Raw Data to Predictions

01

Load

Iris dataset

→

02

Split

80 / 20 stratified

→

03

Scale

StandardScaler

→

04

Train

KNN, k=5

→

05

Evaluate

Metrics + matrix

Design principle: keep preprocessing inside the model pipeline so the same transformation is applied consistently at training and inference time.

Scaled K-Nearest Neighbors

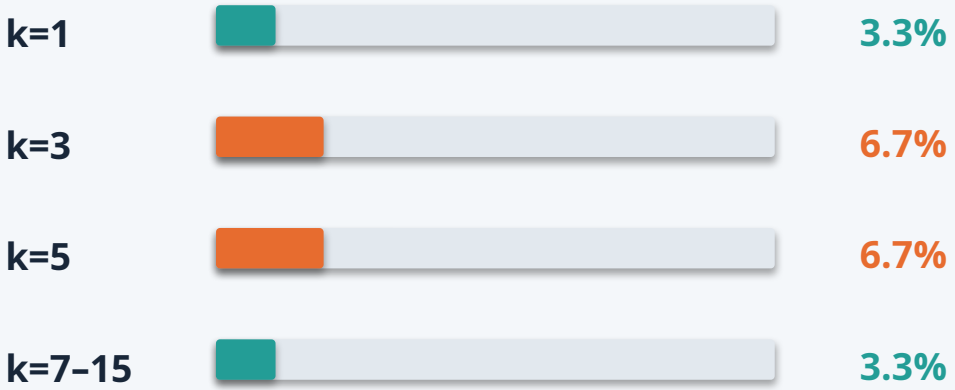
WHY KNN?

Similar flowers tend to occupy nearby points in feature space.

- Simple and interpretable
- Strong fit for a small numerical dataset
- Scaling prevents large-unit features from dominating

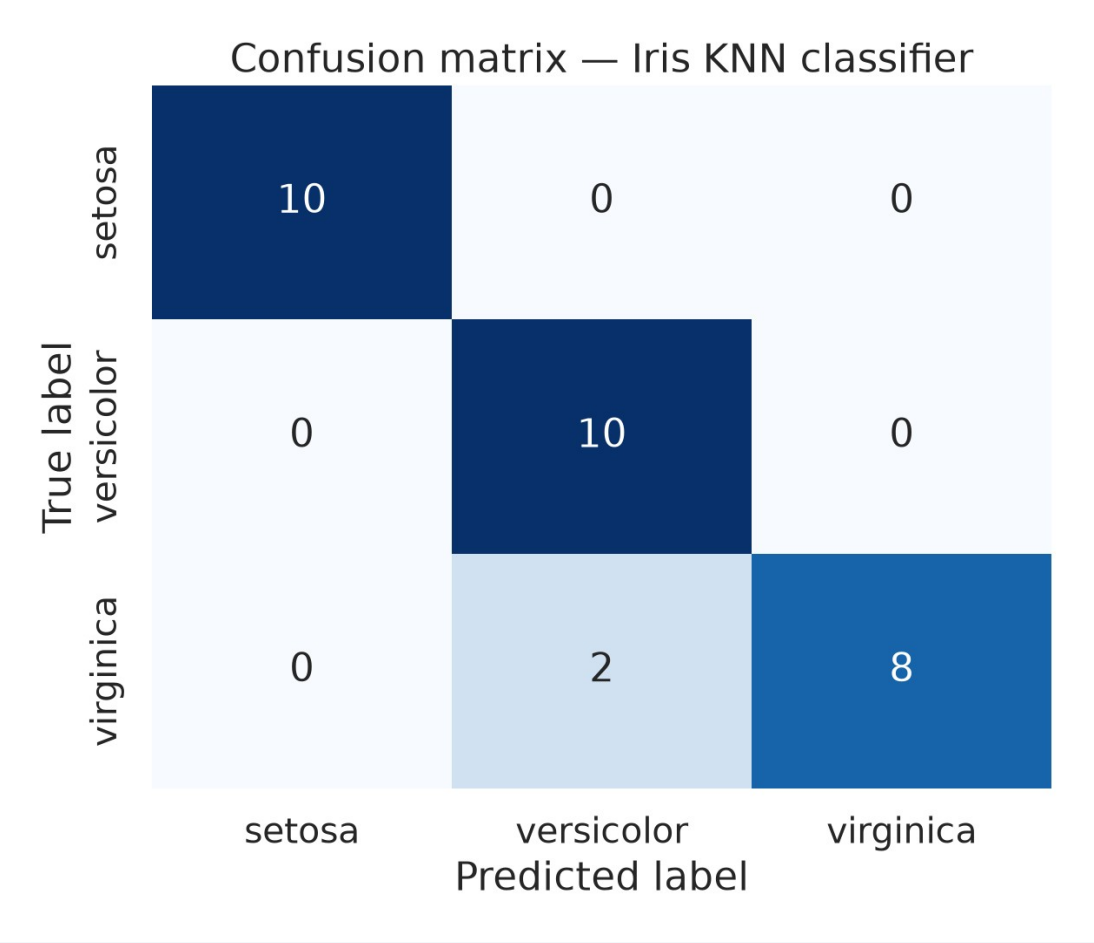
TUNING K

The project tests odd k values from 1 to 15.



Final configuration: StandardScaler → KNN(k=5)

Evaluation: What Did the Model Miss?



READING THE MATRIX

Setosa was classified perfectly.

The only errors were two Virginica flowers predicted as Versicolor. This is the expected harder boundary between these visually similar classes.

28 / 30 test samples correct

Source: executed project artifact · confusion_matrix.png

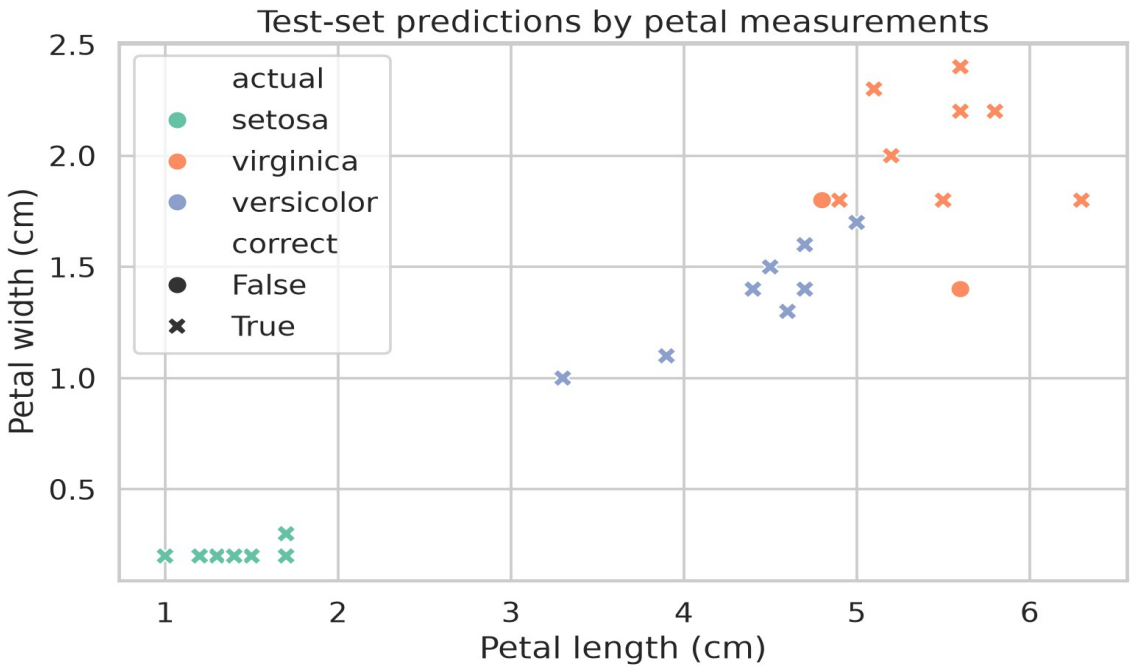
Results & Evidence

93.33%
ACCURACY

94.44%
PRECISION

93.33%
RECALL

93.27%
WEIGHTED F1



WHAT THIS DEMONSTRATES

- Reproducible data preparation
- Distance-aware model selection
- Multi-metric model validation
- Clear error interpretation

Source: executed project artifacts · metrics.json and prediction_scatter.png

07 · CLOSE

A small model, built like a professional system.

This project bridges theory and practice: data understanding, supervised learning, reproducible evaluation, and communication of results.

NEXT PRODUCTION STEPS

- Cross-validation and experiment tracking
- API deployment and monitoring
- Data drift and model governance

Thank you

Muhammad Bello Abdussalam